

Level I — Full condensate description
One globally unitary Euclidean condensate
No independent quantised metric field
No fundamental collapse
Non-factorisable correlations intrinsic

partial tracing

Reduction map $\rho = \text{Tr}_{\text{env}} |\Psi\rangle \langle \Psi|$
(partial trace over unresolved condensate modes)

$\Gamma_{\text{loop}} \ll 1$

$\Gamma_{\text{loop}} \gg 1$

Level IIa Reduced coherent-effective regime
 $\Gamma_{\text{loop}} \ll 1$ (mesoscopic)
Coherent common-medium structure intact
Interference survives · Reduced, not classical

Level IIb Reduced macroscopic classical-looking
 $\Gamma_{\text{loop}} \gg 1$ (macroscopic)
Strong locking + decoherence
Effective Newton / Einstein gravity · Apparent
irreversibility, reduced thermality, stochasticity

GIE structurally allowed
Correlated common-medium loophole class
(Trillo–Navascués-type route)
No violation of reduced-state logic

Standard no-go logic applies
Local classical factorising channel
No gravity-induced entanglement
BMV no-go applies straightforwardly

Architectural caveat. “Classicality” in ECT is not a separate ontological sector but a regime of reduced description. Denying a fundamental quantised metric is *not* denying the quantum character of the underlying medium.